

CARF: Causal Reasoning Augmented Reflection for Long-Term Planning via LLMs

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Abstract—Recent developments suggest that Large Language Models (LLMs) can identify and correct errors in their generated responses using reflection mechanisms. However, when applied to long-term task planning, these methods reveal significant limitations. Reflection methods may neglect the causes of errors in earlier planning, producing results that contain inaccurate information, thereby leading to further mistakes in subsequent planning. This paper explores the reflection frameworks suitable for long-term task planning. Inspired by human causal cognitive processes, we introduce the Causal Reasoning Augmented Reflection Framework (CARF). CARF employs systematic causal reasoning to accurately identify the root causes of errors and to generate effective action plan revisions by integrating association information. We conducted household task experiments in *AlfWorld* and *VirtualHome*, and the results show that our framework significantly improves the success rate of complex long-term tasks. Furthermore, we applied CARF to a real-world robotic system, highlighting its potential for practical applications. Website at <https://hutcher.github.io/CARF.github.io/>

I. INTRODUCTION

Task planning is a critical decision-making process widely applied in fields such as robotics navigation [1], manipulation [2], [3], and everyday household tasks [4]–[8]. While research on short-term decision-making has progressed significantly, the complexity and error accumulation inherent in long-term task planning continue to present substantial challenges. Long-term task planning requires the management of extended action sequences and more complex environmental states. Additionally, in long-term planning, the issue of error accumulation becomes more evident, as initial mistakes may cause progressively greater deviations over time.

Recently, large language models (LLMs) have been incorporated into task planning due to their strong common-sense reasoning and logical capabilities, as seen in methods like ReAct [9] and ADAPT [10]. However, these LLM-based methods can face challenges like hallucinations and contextually unfaithful responses [11]–[13]. To address these issues, recent studies have introduced reflection mechanisms into task planning, employing either human or automatic feedback to refine erroneous outputs. This transition from an open-loop to a closed-loop system is exemplified by methods like RETROFORMER [14] and Reflexion [15]. Therefore, we pose the question: Can the LLM-based reflection framework be applied to long-term task planning?

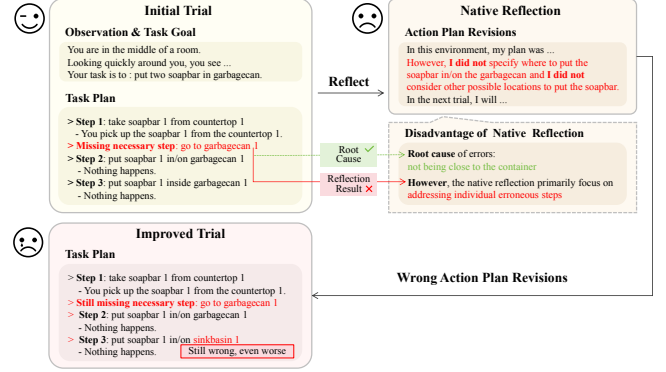


Fig. 1. Illustrates the limitation in using reflection methods for long-term task planning. These methods merely identify and correct one or a few erroneous steps, overlooking the root causes of errors. However, correcting only these steps may lead to greater errors in subsequent trials.

Although the above methods have achieved some initial success, a key challenge in applying reflection methods to long-term task planning lies in their inability to identify the root causes of errors (RCE), which is essential for the subsequent generation of action plan revisions. Presently, these reflection methods do not concentrate on understanding the reasons behind errors in long-term task planning. Instead, they focus solely on revising the action plan by addressing only part of the erroneous steps [14]–[16]. For example, in Figure 1, reflection methods focus on testing different placement strategies for a soapbar instead of investigating why the failure occurred in the first place. This neglect of identifying the RCE leads to subsequent trials that are even more erroneous. In some cases, it can even result in dangerous actions that compromise the robot’s reliability.

Motivated by the above challenges, this paper investigates the analysis of RCE in complex long-term task planning. Such planning typically involves extended action sequences and accumulates errors over time. Therefore, simply identifying and correcting one or a few erroneous steps does not fully rectify the errors in the planning. Only by accurately identifying the RCE can one generate action plan revisions that address the root causes, resolving the accumulating errors and successfully completing the task in subsequent trials. However, current reflection methods predominantly focus on addressing individual erroneous steps to generate action plan revisions. In contrast, human intelligence is characterized by the ability to perform causal reasoning across three levels: Counterfactuals, Association, and Intervention [17]–[19]. These levels range from simple observations to complex

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imaginings, enabling humans to systematically analyze the RCE and propose effective action plan revisions, thus enhancing the success rate of subsequent trials.

Inspired by the process outlined above, we propose CARF, a general reflection framework that harnesses the causal reasoning capabilities of LLMs to analyze the RCE and generate effective action plan revisions for future trials. First, for the failed action plan, we utilize counterfactual reasoning to substitute critical erroneous steps from previous trials and identify possible the RCE, corresponding to the third level of causal reasoning. Next, we employ LLMs to infer associated information from environmental observations, corresponding to the first level of causal reasoning. Finally, based on the task trajectory, associated information, and the RCE in the previous action plan, we formulate action plan revisions aimed at addressing the root causes and successfully complete the next trial, corresponding to the second level of causal reasoning. To further enhance the reflection optimization process, we integrate memory management to improve the LLM’s performance in subsequent trials.

To assess the effectiveness of the proposed framework, we conducted experiments in Alfworld and categorized the dataset into three task types. The results show that our proposed framework significantly outperforms the baseline, achieving a 34% higher success rate across all tasks compared to the Planning-only method. Furthermore, applying CARF to the VirtualHome environment improves the success rate on the NovelTask dataset by 27% over the Planning-only approach. Finally, we deployed our framework on a real-world robotic system for rearrangement tasks, demonstrating its potential for real-world applications. In summary, the contributions of this paper are as follows:

- 1) To our knowledge, we are the first to study the reflection frameworks in long-term task planning.
- 2) We have introduced a framework named CARF. Our framework effectively enhances the application of reflection frameworks in long-term task planning by identifying the RCE, obtaining association information, and then generating action plan revisions.
- 3) We conducted experiments in the virtual household environments Alfworld and VirtualHome, as well as in real-world tasks. The results demonstrate that our framework could effectively enhance the success rate of complex long-term task planning.

II. RELATED WORKS

A. LLMs for Task Planning

In recent years, the general performance exhibited by LLMs has sparked interest among researchers in exploring the potential of LLMs for task planning in open-ended environments. For instance, LID [20] uses GPT-2 as a policy network, predicting the next action based on encoded tasks information. ReAct [9] introduces a new prompting paradigm that combines reasoning and action to enhance the ability of LLMs to solve general tasks. CLIN [21] features persistent dynamic text memory centered on causal abstraction,

enabling LLM-based agents to continually learn and adapt in new environments. RAP [22] combines a retrieval-based mechanism with a generative model, boosting multimodal LLM agents’ planning capabilities. SYNAPSE [23] employs a trajectory-based prompting mechanism that selects and uses similar past trajectories from memory.

B. LLMs for Reflection

To address the challenges of hallucinations or context unfaithfulness in outputs from LLMs, researchers have investigated the application of reflection mechanisms to improve the quality of outputs. For example, SELF-REFINE [24] is an iterative self-improvement framework that has shown significant improvements in single-step reasoning tasks by providing feedback on the current output and refining it based on the feedback. Reflexion [15] introduces a framework that uses language feedback as a semantic gradient signal. CRITIC [25] interacts with specific tools to assess and correct errors and flaws in the generated text. SelfCheck [26] corrects mistakes by regenerating specific steps in the reasoning process and comparing them with the original steps.

C. LLMs for Causal Reasoning

One potential solution to the aforementioned problem is to leverage causal reasoning to identify the RCE. In complex long-term tasks, errors accumulate over time, and causal reasoning can pinpoint the RCE, enabling the development of effective action plan revisions. Recent research has shown that LLMs have potential in causal reasoning, as they can outperform existing causal reasoning algorithms by utilizing common sense and domain knowledge in tasks such as counterfactual reasoning and real-world causal relationships [17], [27]. Furthermore, even in the absence of specific domain knowledge, LLMs can still perform limited causal reasoning by leveraging the available data [28].

For complex long-term tasks, if reflection frameworks fail to identify RCE, it can lead to ineffective or incorrect plan revisions, which in turn may cause future trials to fail again. Applying reflection frameworks to long-term task planning remains a challenging and unresolved issue.

III. PRELIMINARIES

A. Planning Framework

A task can be defined as a tuple $\langle S, O, A, T \rangle$, where S represents the set of states, O is the set of observations, A denotes the set of actions, and T describes the stochastic transitions [29]. The transition function, formalized as $T: S \times A \rightarrow S$, represents the environmental state changes resulting from actions taken. The objective is to determine a sequence of actions that transitions from the initial state to the target state, thereby forming a plan. Although there is no strict definition for long-term tasks, they generally involve 5 to 15 or more steps and necessitate extensive interactions with objects and the environment, in contrast to short-term tasks.

In the reflection method, the sequence of actions is generated by the strategy function. It can be defined as $\Phi(a_i^t | g^t, h_i^t, o_i^t, C^t)$, where determines the next action $a_i^t \in$

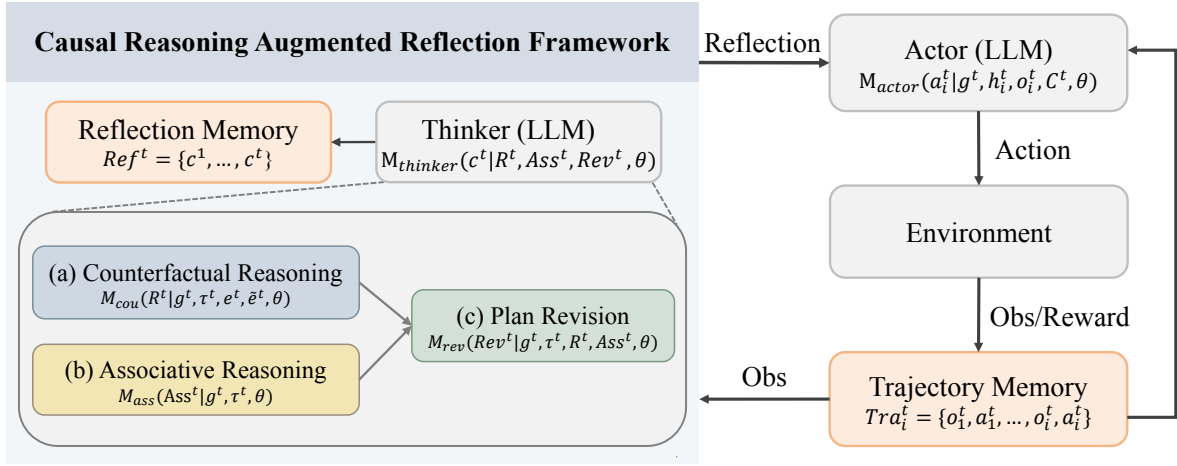


Fig. 2. The framework of CARF. Our framework operates by taking historical task information and environmental feedback as input and producing reflection results as output. The framework comprises three stages: 1. Counterfactual Reasoning is responsible for identifying and correcting critical erroneous steps and analyzing the RCE. 2. Associative Reasoning is tasked with inferring about the associations. 3. Plan Revision is charged with generating the action plan revisions for subsequent trials. Additionally, we have introduced a Memory Management module to manage the historical results generated.

A , based on the task goal g^t , the historical information $h_i^t = \{o_1^t, a_1^t, \dots, o_{i-1}^t, a_{i-1}^t\}$, the observation $o_i^t \in O$, and the historical reflection information set C^t . Here, t is the index of the trial, and i is the index of the step within the t -th trial.

IV. METHODOLOGY

In this section, we will provide a detailed introduction to our proposed framework, CARF, and its interactions with external systems, as shown in Figure 2. The external system primarily includes the strategy model and the environment. We define the strategy model as the Actor, which uses task goal, historical information, environmental observations, and reflection results as inputs to generate actions for interacting with the environment. Our reflection framework is defined as the Thinker, which integrates task information and environmental feedback to conduct causal reasoning. After a trial, the Thinker initially utilizes counterfactual reasoning to identify potential RCE. Subsequently, it applies associative reasoning to infer association information from environmental observations. Finally, the Thinker combines the RCE with the association information to generate action plan revisions. Additionally, we have introduced a Memory Management module to manage the historical results generated.

A. LLMs as Actor

Following [15], we use the React method [9] as the Actor module *Actor*, due to its outstanding performance in task planning. As an implementation of the strategy function, M_{Actor} takes information about the task and environment as inputs and outputs the next action. It can be represented as:

$$a_i^t = M_{actor}(g^t, h_i^t, o_i^t, C^t, \theta), \quad (1)$$

where a_i^t denotes the next action output, g represents the task goal, h_i^t encapsulates the historical information, o_i^t indicates the environmental observations, $C^t = \{c^{t-k}, \dots, c^{t-1}\}$ represents the set of reflection outcomes from the past k trials,

and θ represents the parameters of LLMs. Additionally, the action and observation at each step will be collected by the memory management module. The Actor module possesses strong natural language understanding and reasoning capabilities, enabling it to adapt flexibly to complex tasks and dynamically optimize execution.

B. LLMs as Thinker

To identify the RCE in previous trials and to generate action plan revisions, we designed the Thinker module, $M_{thinker}$. As illustrated in Figure 3, The Thinker consists of three submodules: The Counterfactual Reasoning submodule, responsible for identifying and correcting critical erroneous steps and analyzing the RCE R^t ; the Associative Reasoning submodule, tasked with inferring the relationships between objects and containers Ass^t ; and the Plan Revision submodule, which generates corrections for the action plan for subsequent trials Rev^t . It can be formalized as follows:

$$c^t = M_{thinker}(R^t, ASS^t, Rev^t), \quad (2)$$

where c^t represents the collection of outputs from each submodule. Furthermore, the results from the Thinker module will be collected by the memory management module.

Counterfactual Reasoning M_{cou} analyzes the RCE by identifying and correcting critical erroneous steps in the original trajectory and evaluating potential outcomes that were not actually executed. During the trial t , M_{cou} takes the task information, and execution results as inputs. It then reasons under the assumption of replacing the actual execution results with counterfactual outcomes, thereby outputting the RCE. This process can be formally expressed as:

$$R^t = M_{cou}(g^t, \tau^t, e^t, \tilde{e}^t, \theta), \quad (3)$$

where R^t represents the RCE for natural language expression, τ^t represents the trajectory, while e^t and $\tilde{e}^t$ respectively

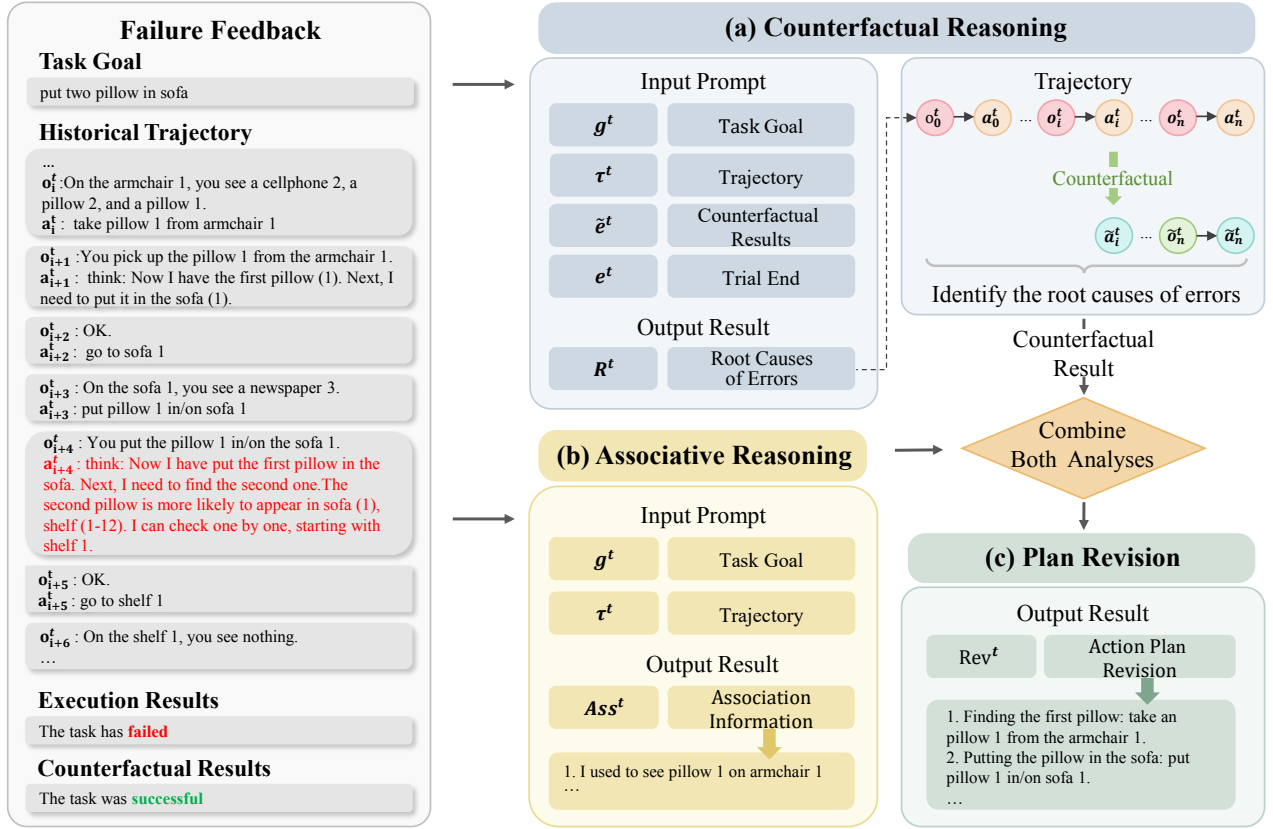


Fig. 3. The detailed illustration of CARF. This framework integrates task information and environmental feedback as input and outputs the RCE in the planning process, the association information, and the action plan revisions

represent the actual execution results and the counterfactual execution results. This submodule, through counterfactual reasoning, can efficiently identify the RCE.

Associative Reasoning M_{ass} infers associations between objects and containers based on observations within the trajectory. During the trial t , M_{ass} takes the task goal and trajectory as inputs and outputs the associations between objects and containers. To minimize redundancy and shorten the context length, we focus only on objects directly related to the task goal. It can be expressed as:

$$Ass^t = M_{ass}(g^t, \tau^t, \theta), \quad (4)$$

where Ass^t represents the association information between objects and containers as specified in the task goal. This submodule, through associative reasoning, can precisely identify the locations of objects, thereby improving search efficiency.

Plan Revision M_{rev} proposes action plan revisions based on the results of counterfactual and associative reasoning. During the trial t , M_{rev} takes the identified RCE, the association information between objects and containers, and the task information as inputs, and outputs the action plan revisions. The process is outlined as follows:

$$Rev^t = M_{rev}(g^t, \tau^t, R^t, Ass^t, \theta), \quad (5)$$

where Rev^t represents the action plan revisions for natural language expression. This submodule, by integrating the

analysis results, generates optimized action plan revisions, significantly improving the success rate of tasks.

The Thinker consists of three submodules, each corresponding to a level of causal reasoning. This structure can efficiently identify RCE, infers association information, and then generates effective action plan revisions, thereby increasing the success rate of long-term tasks.

C. Memory Management

Previous studies frequently incorporated all essential information into the prompt [30]. However, for long-term tasks, this method could result in overly long prompts, taxing the memory capacities of LLMs. To address this challenge, we designed a Memory Management module mem to manage the trajectories and the analysis results from the Thinker module during the planning process. The trajectory management component is denoted as Tra , which manages trajectory information to generate subsequent actions. It is defined as $Tra_i^t = \{o_1^t, a_1^t, \dots, o_i^t, a_i^t\}$, where Tra_i^t represents the set of actions and observations up to step i of the trial t . The reflection management component is represented as Ref , which manages the reflection results from the Thinker to optimize the next trial. It is defined as $Ref^t = \{c^1, \dots, c^t\}$, where Ref^t represents the accumulated set of reflection results up to the trial t . The introduction of the Memory Management module alleviates the pressure on LLMs from

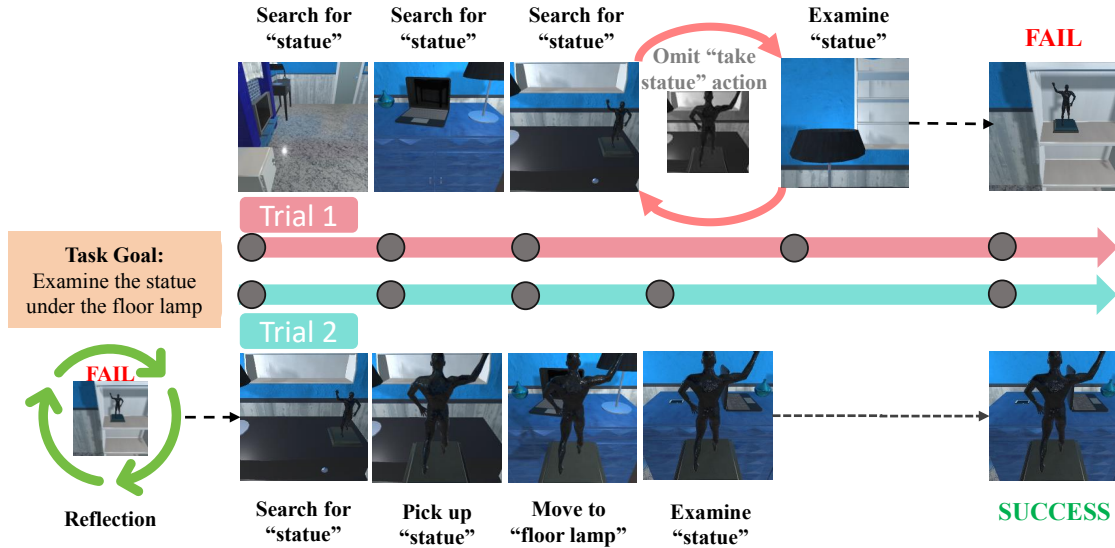


Fig. 4. Example of a failed task successfully replanned using our framework. In the first trial, the agent entered an error loop, repeatedly failing to examine the statue and then searching for it to correct the error, while neglecting to pick it up. In the second trial, our framework identified the root cause of error and generated an optimized action plan revisions, successfully completing the task.

TABLE I

OVERALL PERFORMANCE CARF AND BASELINES ACROSS VARIOUS TASKS. OUR FRAMEWORK OUTPERFORMS ALL BASELINES. IDENTIFICATION AND REVISION ARE SOLELY USED TO EVALUATE THE REFLECTION METHOD.

Categories	Methods	Pick Tasks			Pick & Act Tasks			Examine Tasks		
		Identification	Revision	Success Rate	Identification	Revision	Success Rate	Identification	Revision	Success Rate
w/o reflection	Planning-Only	-	-	56	-	-	54	-	-	18
	React-Only	-	-	81	-	-	73	-	-	82
With reflection	React-Reflexion	68	62	84	68	61	80	40	60	73
	React-CARF	73	70	85	77	74	85	100	100	100

handling overly lengthy prompts, thereby significantly improving the planning process.

V. EXPERIMENTS

In this section, our framework is evaluated through experiments conducted in a virtual household environment.

A. Experimental Setup

Environment. we conducted five trials to assess different methods in Alworld, a semantically-based virtual environment used for training and assessing intelligence. Alworld consists of 120 rooms, each dynamically populated with a set of portable objects and static containers. Intelligent agents can interact with these objects and containers to perform everyday tasks such as cooling, cleaning, and heating. The diversity of task environments and goals provides a comprehensive test for intelligent agents.

Dataset. We conducted experimental tests on the unseen dataset provided by [31]. The unseen dataset includes six tasks: Pick & Place, Examine in Light, Clean & Place, Heat & Place, Cool & Place, and Pick Two & Place. We performed experiments on 100 tasks to obtain results. To minimize the impact of the number of tasks, we categorized them based on task complexity and type into three categories: Pick Task,

Pick & Act Task, and Examine Task, with each category averaging around ten steps.

Compare methods. We compared our method with three previous approaches: 1. **Planning-Only:** ProgPrompt [30], which directly generates the next action by inputting a simple context containing the task description into the LLM; 2. **React-Only:** React [9], which incorporates a reasoning process during planning to derive the next action based on reasoning; 3. **React-Reflexion:** Reflection [15], which integrates a reflection process during multiple trials in React, obtaining the next action based on the reflection from the previous trial. Among all the comparison methods, we use the GPT-4 series as the backbone LLM.

Metrics. We evaluated the performance from three aspects, with the first two metrics specifically targeting the reflection method: **Identification** assesses whether the reflection method can identify the RCE by counting whether each task’s reflection identifies the RCE; **Revision** evaluates whether the reflection method revises errors from previous plans by counting whether each task’s reflection has revised these errors; **Success Rate** measures the effectiveness of the reflection and planning methods by counting whether each task is successfully completed [20], [32].

B. Results

The primary results are presented in Table I. We can observe that: 1. our framework consistently outperforms other methods across all metrics and task categories. This indicates that our framework can efficiently identify the RCE, thereby generating effective action plan revisions and enhancing the success rate. Figure 4 exemplifies the application of CARF to long-term tasks. 2. A noticeable trend is that a low success rate corresponds to a low rate of identifying the RCE. This highlights a major issue in long-term task planning: a low identification of the root causes leads to inaccuracies in action plan revisions, severely impacting the successful execution of subsequent plans. 3. The **React-only** method and the **React-reflexion** method exhibit similar performance across various tasks, indicating that reflection methods that only revises action plans for one or a few erroneous steps are not effectively resolving errors encountered in long-term task planning. 4. The **React-CARF** method demonstrates superior performance compared to methods like **React-reflexion**, indicating that CARF can pinpoint the RCE, thereby effectively enhancing task success rate. Furthermore, the differences in identification, revision, and success rate across different task categories suggest that the effectiveness of reflection methods may be influenced by task complexity.

VI. ANALYSIS AND DISCUSSION

A. Episode Analysis

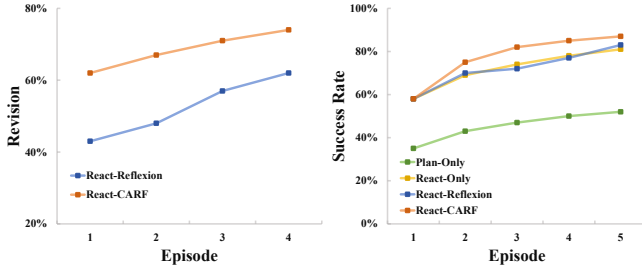


Fig. 5. Result of revision and success rate with different episodes. CARF effectively improves the performance of long-term tasks.

To analyze the effectiveness of methods, we conducted an episode analysis, with results displayed in Figure 5. Our observations are as follows: 1. Our framework consistently outperformed the baseline across all episodes, demonstrating its effectiveness in improving revision and success rate for long-term tasks. 2. Across all episodes, the React-only method consistently outperforms the Planning-only approach. However, the React-Reflection method does not exceed the performance of the React-only method. This further confirms our previous assertion that without identifying the RCE in planning, reflection methods that merely revises action plans for one or a few erroneous steps cannot achieve the goal of optimizing subsequent trials. These results highlight the importance of integrating causal reasoning into the reflection process, as our framework’s ability to accurately identify root causes and generate effective plan revisions significantly enhances success rate in long-term task planning.

B. Method Performance Across Different LLMs

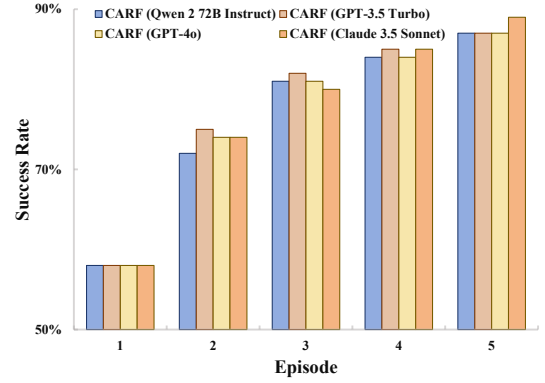


Fig. 6. Result of success rate with different LLMs. CARF possesses outstanding universality and robustness.

To further investigate the universality and robustness of our framework, we conducted experiments with CARF on multiple LLMs of varying scales and architectures. The results, as shown in Figure 6, indicate the following: 1. Across all LLMs, our framework consistently outperforms the baseline, demonstrating outstanding universality. 2. Our framework displays similar performance across LLMs with different architectures, indicating remarkable robustness. Additionally, based on different LLMs, our framework exhibits varying growth rates across episodes, highlighting the distinct advantages each LLM brings to handling complex tasks. The framework effectively utilizes these model characteristics to achieve dynamic adaptability. These results underscore the wide applicability of our framework, enabling sustained optimization of performance across various LLMs and addressing the demands of complex and variable tasks.

C. Method Performance on the VirtualHome Benchmark

TABLE II
PERFORMANCE COMPARISON IN VIRTUALHOME. OUR FRAMEWORK OUTPERFORMS ALL BASELINE MODELS.

	Success Rate
Planning-Only	50
React-Only	53
React-Reflexion	70
React-CARF	77

We conducted additional evaluations in VirtualHome, a 3D residential environment featuring various homes and objects. In this setting, the agent must make multi-step decisions and manipulate objects to complete household tasks. To assess performance, we performed comparative experiments on 100 tasks from the NovelTask dataset provided by LID [20].

To further verify the generalizability of our method across different environments, we conducted experiments in VirtualHome. The results, presented in Table II, reveal the following key findings: 1. Our method achieves a task success rate of 77% in the VirtualHome environment, significantly outperforming all baseline models. 2. Compared to the React-Only

Task: Place the water bottles from the initial area in a row within the target area.

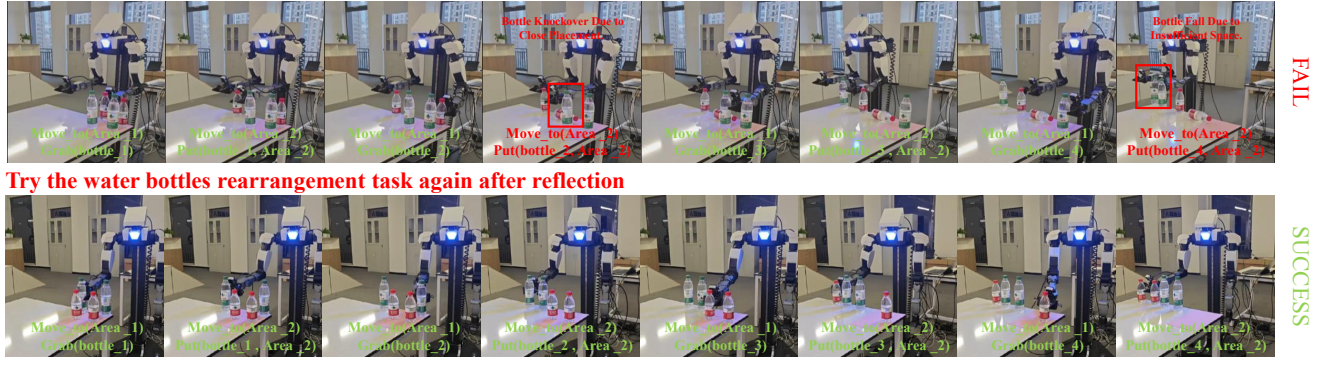


Fig. 7. Example of Successful Replanning After Initial Failure in Water Bottles Rearranging. In the first trial, two errors occurred: the second bottle was placed too close to the first, causing it to topple, and the third bottle was positioned too far from the second, resulting in the fourth bottle falling. In the second trial, our framework identified and revised these errors, successfully completing the bottle rearrangement task.

method, the React-Reflexion method increases the success rate by 17%, validating the effectiveness of reflection in task execution. 3. Compared to the React-Reflexion method, our approach further optimizes the reflection process, achieving a 24% higher success rate than the React-Only method, highlighting its advantage in long-term task planning.

D. Ablation Study

TABLE III

ABLATION OF VARIOUS SUBMODULES OF CARF. ALL DESIGN SUBMODULES CONTRIBUTE TO THE PERFORMANCE. WITHOUT THE COUNTERFACTUAL SUBMODULE, IDENTIFICATION CANNOT BE MEASURED.

	Identification	Revision	Success Rate
Ours full	76	74	87
w/o Associative	60	67	78
w/o Counterfactual	-	67	79
w/o Both Analyses	-	52	75

To comprehensively demonstrate the effectiveness of each submodule in our framework, we conducted ablation studies. The results are detailed in Table III. In the **w/o Associative** model, action plan revisions are generated without association information. Compared to our complete model, the three metrics respectively decreased by 16%, 7%, and 9%. The results indicate that association information contributes to improving identification, which in turn enhances success rate. This is linked to the key role of association information in forming effective action plan revisions, which bring trajectories closer to the correct paths and reduce the difficulty of identification. In the **w/o Counterfactual** model, action plan revisions are generated without identifying the RCE. Compared to our complete model, the w/o Counterfactual shows a 7% decrease in revision, leading to an 8% decline in success rate. This suggests that utilizing the counterfactual reasoning submodule to identify the RCE can significantly enhance the revision and thereby increase success rate. The **w/o Both Analyses** model directly generates action plan revisions. Compared to our complete model, this variant's

revision decreased by 22%, and the success rate dropped by 12%. These results highlight the critical roles of the associative and counterfactual reasoning submodules in improving revision and success rate.

VII. REAL-WORLD APPLICATION

To illustrate the practical potential of our framework, we developed a CARF-based planning system to demonstrate its implementation in intelligent systems.

We evaluated our system on real-world long-term tasks, such as rearranging water bottles, using four atomic actions: `Move_to(position)`, `Grab(object)`, `Put(object, position)`, and `Take_picture()`. The system was tested under various conditions. In simpler scenarios, the reflection method only needed to identify the cause of a single-step failure. In more complex cases, it analyzed the impact of previous steps on the current step to determine the root cause of failure and correct errors.

As shown in Figure 7, our framework effectively analyzes both single-step failures and interdependencies between multiple steps, enabling precise action plan revisions. These results demonstrate the strong performance of our approach and highlight its practical applicability in real-world tasks, offering promising prospects for robotic applications.

VIII. CONCLUSIONS

To our knowledge, we are the first to investigate the application of reflection frameworks in long-term task planning. Inspired by human intelligence, we propose a new framework named CARF, which integrates three submodules corresponding to the three levels of causal reasoning to enhance reflection processes. Additionally, we introduce a memory management module to efficiently handle the context of LLMs. Experimental results from both virtual household environments and real-world scenarios demonstrate that our framework significantly enhances error identification, action plan revisions, and success rates in long-term task planning, highlighting its strong generalizability and robustness.

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